Corporate Finance (in lingua inglese) - A.A. 2025/2026 - Topic 5d and 8: NPV and other investment criteria

TEXT and SOLUTION (through the use of Excel)

Event Corporation is planning to set up a new gift shop. The firm will use a building, from which it currently collects a rent of € 20 a year. The project will require an initial investment of € 1,000 in forniture which will be depreciated over a period of 4 years, using straight-line depreciation. The additional annual sales will be € 900 while the incremental costs are expected to be € 450 per year for the entire period (depreciation is not included). The value of the working capital is likely to increase by 20% in each year, from year 2 to year 3 and, at the end of the fourth year, the working capital will be 0. The working capital in year 1 would be 500. The tax rate is 38%, the cost of debt is 8.3% (firm's debt/assets = 0.7), the risk premium on the firm's stock is 13% and the risk-free rate is 5%. The project duration is 4 years. No salvage value. The risk associated with the project and its indebtedness are comparable to that of the firm as a whole. Calculate NPV and IRR. Verify if the IRR you determined si correct (i.e., NPV = 0).

WACC = re*E/V + rd(1-tc)*D/V = (0,05+0,13)*0,3 + 0,083*(1-0,38)*0,7 = 0,09 = 9%

				Period		
	Year	**0**	**1**	**2**	**3**	**4**
	Panel A Capital Investment					
1	Cash flow from investment in fixed assets	-1000,00				
2	Sales of fixed assets					
3	Less tax on sales					
4	Cash flow from capital investment (1+2+3)	-1000,00				

	Panel B Operating Cash Flow					
5	Revenues		880,00	880,00	880,00	880,00
6	Expenses*		450,00	450,00	450,00	450,00
7	Depreciation		250,00	250,00	250,00	250,00
8	Pretax profit (5 - 6 - 7)		180,00	180,00	180,00	180,00
9	Tax at 38%		68,40	68,40	68,40	68,40
10	Profit after tax (8 - 9)		111,60	111,60	111,60	111,60
11	Operating cash flow (10 + 7)		361,60	361,60	361,60	361,60

	Panel C Investment in Working Capital					
12	Working capital		500,00	600,00	720,00	0,00
13	Change in working capital		500,00	100,00	120,00	-720,00
14	Cash flow from investment in w. c . (-13)		-500,00	-100,00	-120,00	720,00

	Panel D Project Valuation					
15	Total project cash flow (4 + 11 + 14)	-1000,00	-138,40	261,60	241,60	1.081,60
16	Discount factor at 9%	1,00	0,92	0,84	0,77	0,71
17	Discounted cash flows (15x16)	-1000,00	-126,97	220,18	186,56	766,23
18	**NPV (sum of discounted cash flows)**	**46,00**				

* Not including depreciation

Alternative method for calculating the discounted cash flows (by putting all the data in a single formula)

Calculations of the discounted cash flows at 9%					
Period	0	1	2	3	4
Formula	-1000,00/(1+9%)0	-138,40/(1+9%)1	261,60/(1+9%)2	241,60/(1+9%)3	1.081,60/(1+9%)4
Discounted cash flows	- 1.000,00	-126,97	220,18	186,56	766,23

IRR

IRR = **10,31%**

If the IRR is correct, then the NPV at 10.31% will be equal to 0 (in this case it is -0.01 owing to rounding):

Period	0	1	2	3	4
Formula	-1000/(1+10,31%)0	138,40/(1+10,31%)1	261,60/(1+10,31%)2	241,60/(1+10,31%)3	1.081,60/(1+10,31%))4
Present values	- 1.000,00	- 125,46	214,98	179,99	730,48
Net present value (sum of PVs)	-0,01				